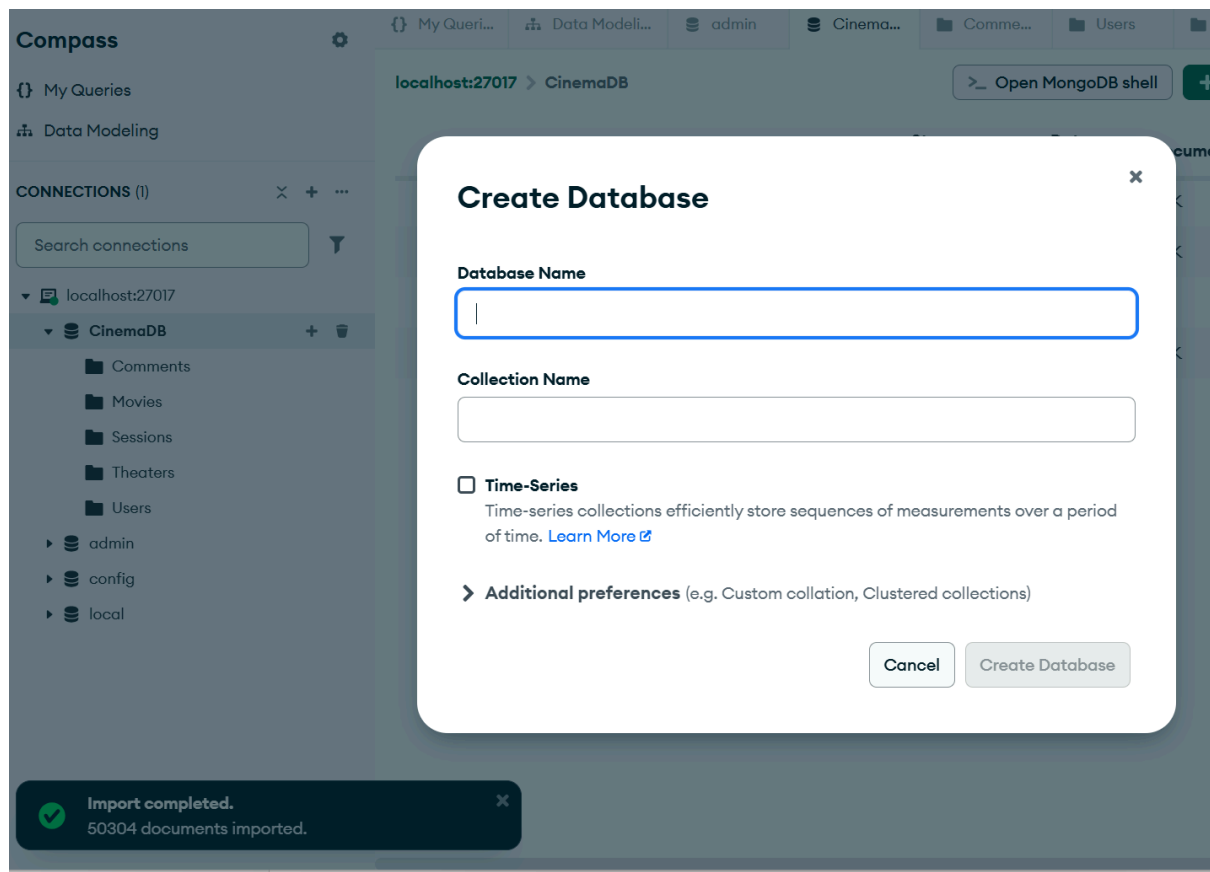
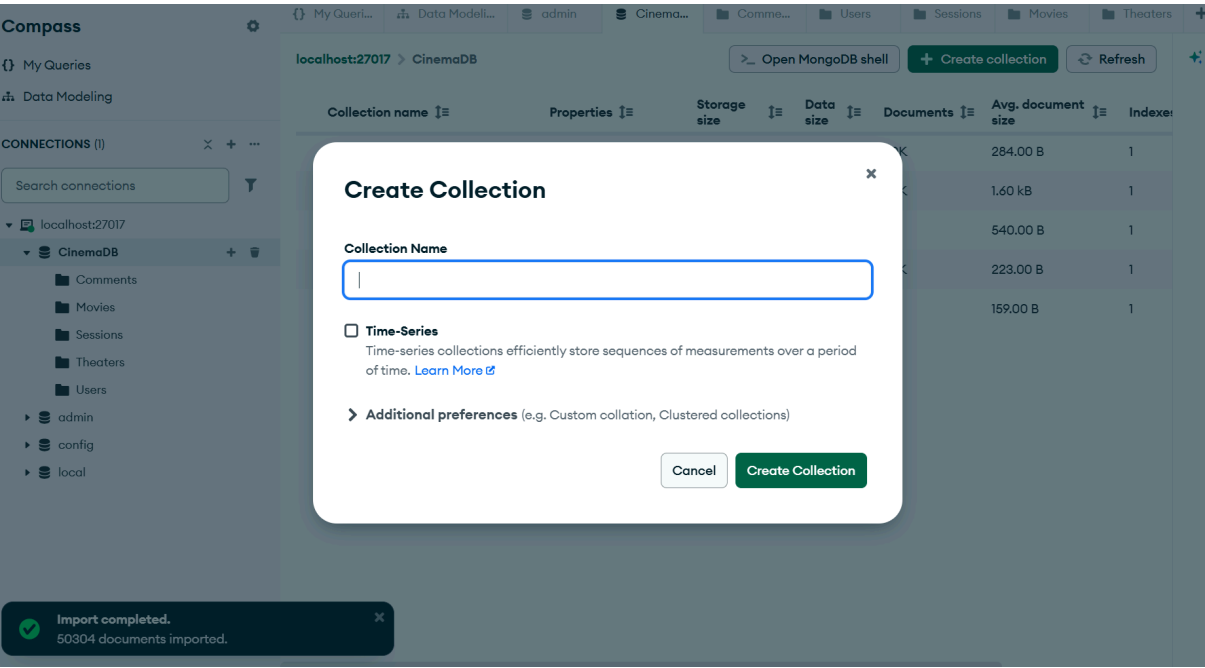


Nivell 1:

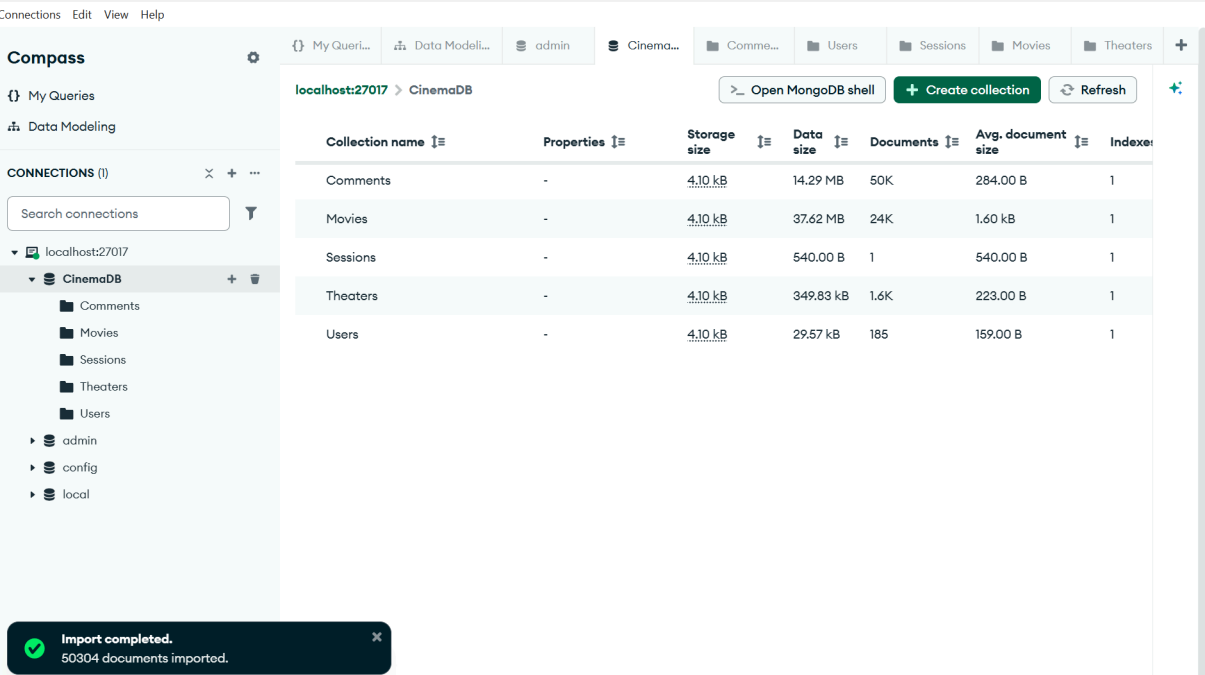
Crea una base de dades amb MongoDB utilitzant com a col·leccions els arxius adjunts.

He creado una base de datos y 5 collections (el equivalente a tablas) desde la UI. Luego he importado los 5 archivos json facilitados en cada collection.





Aquí se puede ver mi base de datos CinemaDB:



Nivell 1 - Exercici 1

1.1 - Mostra els 2 primers comentaris que hi ha en la base de dades.

He generado la query con la ayuda del asistente IA de MongoDB.

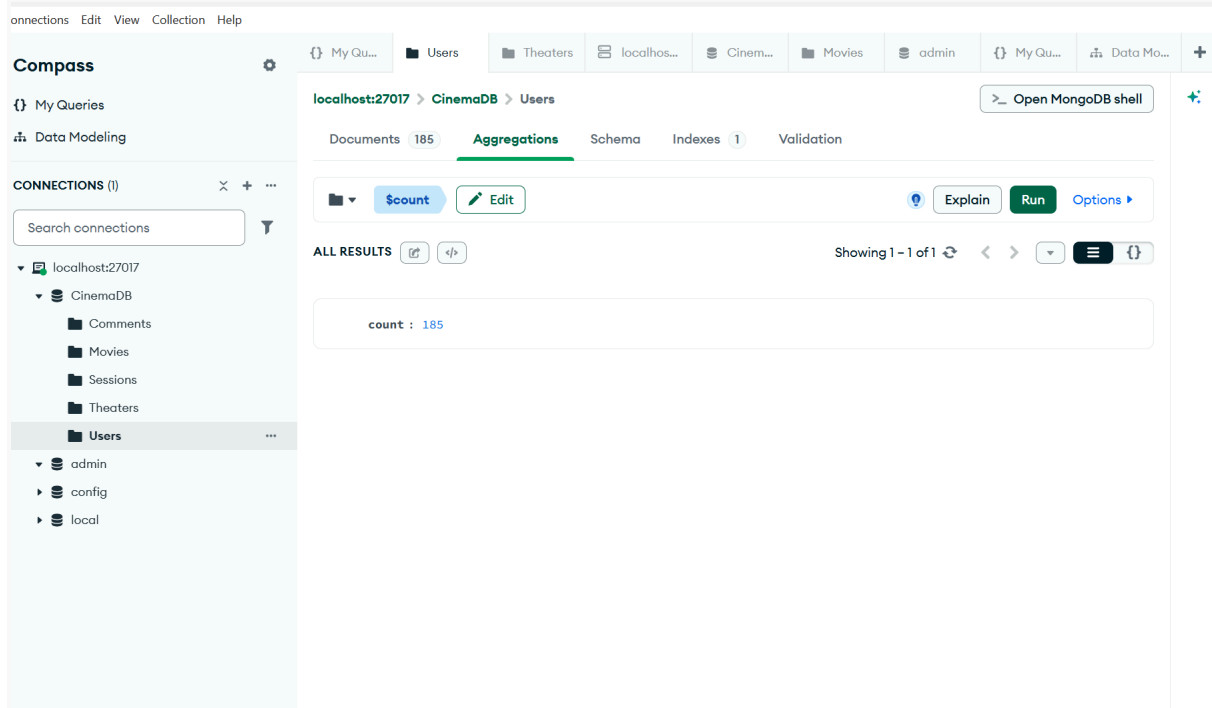
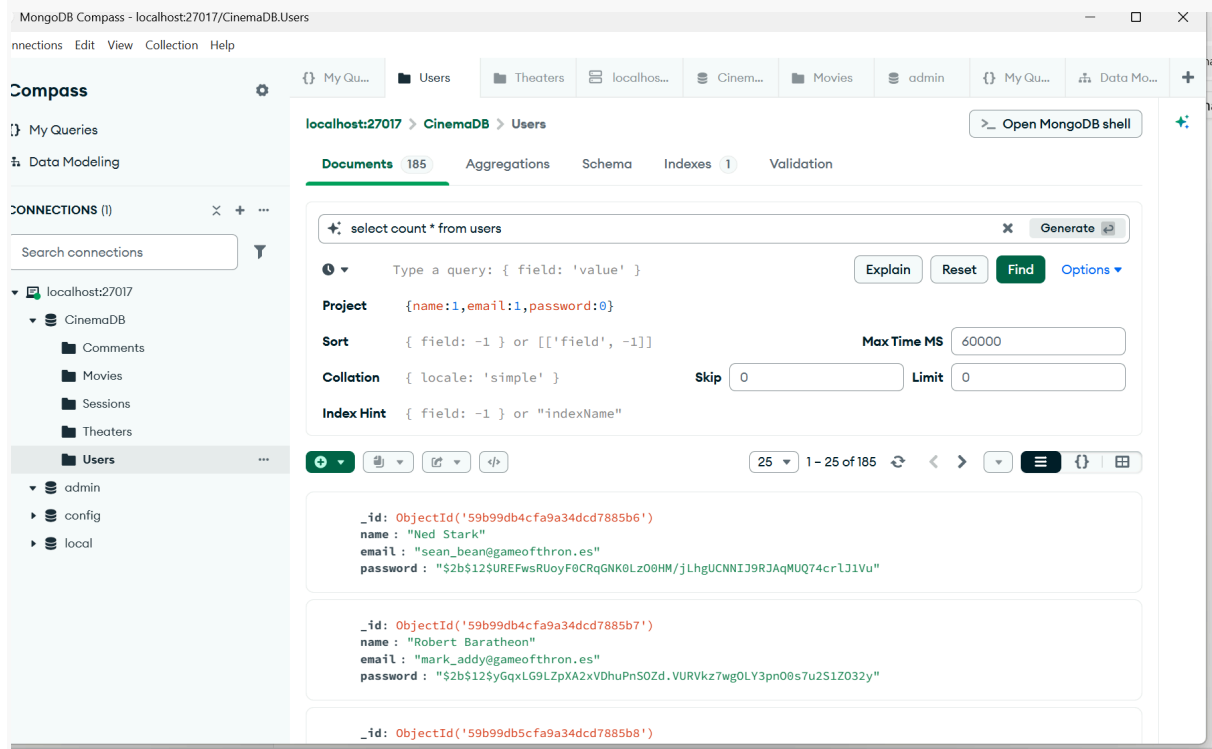
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel shows a connection to 'localhost:27017' with a database named 'CinemaDB'. The 'Comments' collection is selected. The main panel shows the 'Documents' tab with a query for the 'First 2 comments'. The query is: `{ field: 'value' }`. The 'Sort' is set to `{ field: -1 } or [['field', -1]]`. The 'Collation' is `{ locale: 'simple' }`. The 'Index Hint' is `{ field: -1 } or "indexName"`. The 'Skip' is 0 and the 'Limit' is 2. The 'Max Time MS' is 60000. The results show two documents:

```
{
  "email": "andrea_le@fakegmail.com",
  "movie_id": ObjectId('573a1390f29313caabdc418c'),
  "text": "Rem officiis eaque repellendus amet eos doloribus. Porro dolor volupta..",
  "date": ISODate('2012-03-26T23:20:16.000+00:00')
}
```

```
{
  "_id": ObjectId('5a9427648b0beebe69579cf'),
  "name": "Greg Powell",
  "email": "greg_powell@fakegmail.com",
  "movie_id": ObjectId('573a1390f29313caabdc41b1'),
  "text": "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Co..",
  "date": ISODate('1987-02-10T00:29:36.000+00:00')
}
```

1.2 - Quants usuaris tenim registrats?

He generado la consulta a partir del código SQL. El resultado es 185.



1.3 - Quants cinemes hi ha en l'estat de Califòrnia?

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the connection 'localhost:27017' with a tree view of databases: 'CinemaDB' (containing 'Comments', 'Movies', 'Sessions', and 'Theaters') and 'Users'. The 'Theaters' collection is selected. The main panel shows the 'Documents' tab for the 'Theaters' collection. A query is entered in the search bar: `select count(*) from Theaters where location is "CA"`. Below the query bar, the filter `{'location.address.state': 'CA'}` is shown. The results are displayed as a list of documents, showing the first four. Each document contains an `_id`, a `theaterId`, and a `location` object. The `location` object contains `address` and `state` fields. The `state` field is 'CA' for all documents shown.

_id	theaterId	location
ObjectId('59a47286cfa9a3a73e51e72e')	1008	{address: {city: 'Los Angeles', state: 'CA'}, state: 'CA'}
ObjectId('59a47286cfa9a3a73e51e734')	1009	{address: {city: 'Los Angeles', state: 'CA'}, state: 'CA'}
ObjectId('59a47286cfa9a3a73e51e737')	1018	{address: {city: 'Los Angeles', state: 'CA'}, state: 'CA'}
ObjectId('59a47286cfa9a3a73e51e739')	1011	{address: {city: 'Los Angeles', state: 'CA'}, state: 'CA'}

1.4 - Quin va ser el primer usuari/ària en registrar-se?

La única collection donde hay fecha es Comments, así que la query la hice ahí.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel shows a connection to 'localhost:27017' with a database named 'CinemaDB'. The 'Comments' collection is selected. The main panel shows the 'Documents' tab with a query filter of 'earliest date'. The query is displayed as: `{ field: 'value' }`. The 'Project' field is `{ field: 0 }`. The 'Sort' field is `{ date: 1 }`. The 'Collation' field is `{ locale: 'simple' }`. The 'Index Hint' field is `{ field: -1 } or "indexName"`. The 'Skip' field is 0 and the 'Limit' field is 1. The 'Max Time MS' field is 60000. The results panel shows one document:

```
{
  "_id": ObjectId('5a9427648b0beeb695e67e'),
  "name": "Mercedes Tyler",
  "email": "mercedes_tyler@fakegmail.com",
  "movie_id": ObjectId('573a1399f29313caabcec3ce'),
  "text": "Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cu...",
  "date": ISODate('1970-01-01T01:07:09.000+00:00')
}
```

1.5 - Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

He escrit el código SQL para generar la query.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'localhost:27017' with a tree view showing 'CinemaDB' > 'Movies'. The main panel is titled 'localhost:27017 > CinemaDB > Movies' and shows the 'Documents' tab with 24K items. A query bar contains the query: `count distinct movies where genre = comedy`. Below the query bar, the JSON query is shown: `{ 'genres': 'Comedy' }`. The results are displayed in a list view, showing the first document for the query. The document is for a movie with the following fields: `_id` (ObjectId), `plot` (string), `genres` (array), `runtime` (integer), `cast` (array), `num_mflix_comments` (integer), `poster` (string), `title` (string), `fullplot` (string), `languages` (array), `released` (ISODate), `directors` (array), `writers` (array), `awards` (object), `lastupdated` (string), `year` (integer), `imdb` (object), `countries` (array), `type` (string), and `tomatoes` (object).

Nivell 1 - Exercici 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

He escrito el código para las 3 condiciones. He encontrado 6 películas que las cumplen.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the local database 'CinemaDB' with collections 'Comments', 'Movies', 'Sessions', 'Theaters', and 'Users'. The 'Movies' collection is selected. The main panel shows the 'Documents' tab for the 'Movies' collection. A query is entered in the search bar: `{[year:1932]([genres:'drama'] or [languages:'French'])}`. The query is executed, and the results are displayed in a table. The first document is shown in the 'JSON' view:

```
{
  "_id": ObjectId("573a1391f29313caabcd9458"),
  "plot": "A young artist draws a face at a canvas on his easel. Suddenly the mou...",
  "runtime": 55,
  "rated": "UNRATED",
  "cast": Array (4),
  "num_mflix_comments": 1,
  "poster": "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWVtYjlmYi00NjA4LTk4ZW...",
  "title": "The Blood of a Poet",
  "lastupdated": "2015-09-16 13:13:05.537000000",
  "languages": Array (1),
  "0": "French",
  "released": ISODate("2010-05-20T00:00:00.000+00:00") May 20, 2010 at 12:00:00 AM UTC,
  "directors": Array (1),
  "writers": Array (1)
}
```


Nivell 1 - Exercici 3

Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

He escrito el código utilizando “and” para que las 3 condiciones se cumplan al mismo tiempo. He utilizado “USA” para countries y me salieron 25 registros - con “United States” no salía ningún resultado.

He comprobado que los registros cumplan con los 3 criterios de búsqueda.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'localhost:27017' with a tree view of databases including 'CinemaDB' and 'Movies'. The main panel is titled 'localhost:27017 > CinemaDB > Movies' and shows 'Documents' (24K). A query is entered in the search bar: `{year:{$gte:2012,$lte:2014} $and:[{awards.wins:{$gte:3,$lte:5}} $and {countries:'USA'}]}`. Below the query, a JSON document is displayed for a movie from 2013, showing details like plot, genres, ratings, and awards (4 wins, 16 nominations). The document also lists countries as 'USA'.

Nivell 2

Exercici 1

Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

Para este ejercicio he tenido que hacer una consulta a través de varias collections. Por eso he generado primero el modelo de datos a través del asistente de MongoDB. He añadido manualmente la relación entre users, comments y sessions.

Al ejecutar la consulta, he encontrado 185 documentos.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'localhost:27017' with a tree view showing databases like 'CinemaDB' and 'admin'. The 'CinemaDB' database is selected, and the 'Users' collection is highlighted. The main panel displays the 'Aggregations' tab for the 'Users' collection. A query is entered in the 'Query Builder' field: `select count * where email contains "GAMEOFTHRON.ES"`. Below the query, the aggregation pipeline is shown with two stages: '\$match' and '\$count'. The 'Preview' tab is active, showing a preview of 185 documents. Two sample documents are displayed: one for 'Ned Stark' and another for 'Robert Baratheon'. The bottom panel shows the aggregation stages, with 'Stage 1' set to '\$match'.

Compass

My Queries Users localhost:27017 CinemaDB Movies admin My Queries Data Modeling

localhost:27017 > CinemaDB > Users

Documents 185 Aggregations Schema Indexes 1 Validation

select count * where email contains "GAMEOFTHRON.ES" Generate

\$match \$count Explain Run Options

Untitled - modified CREATE NEW PREVIEW

185 Documents in the collection

Preview of documents

```
{ "_id": ObjectId('59b99db4cfa9a34dcd7885b6'), "name": "Ned Stark", "email": "sean_bean@gameofthron.es", "password": "$2b$12$UREFwsRUoyF0CRqGNK0Lz08H..." }
```

```
{ "_id": ObjectId('59b99db4cfa9a34dcd7885b7'), "name": "Robert Baratheon", "email": "mark_addy@gameofthron.es", "password": "$2b$12$yGqxLG9LZpXA2xVDhuPnS0Zd..." }
```

Stage 1 \$match

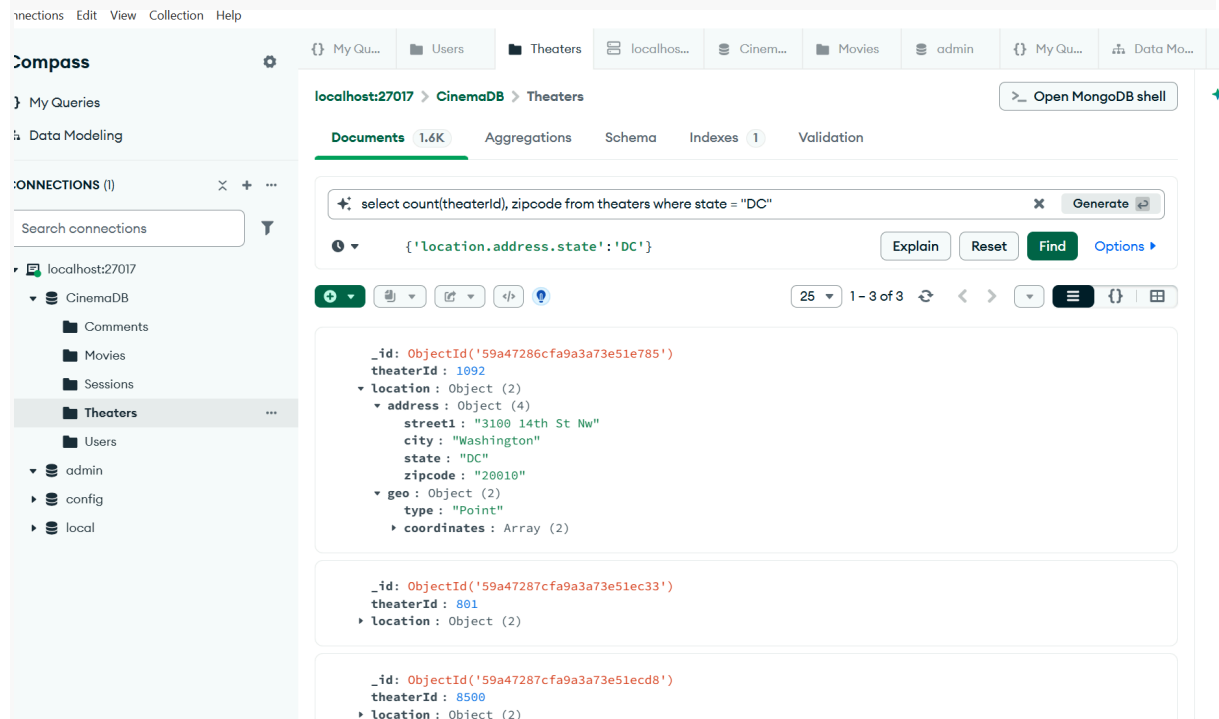
```
1 {
2   email: {
```

Output preview after \$match stage (Sample of 0 documents)

Nivell 2 - Exercici 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

He escrito el código en SQL. He encontrado 3 cinemas - 1 por zipcode - en el estado DC.



The screenshot shows the MongoDB Compass interface. On the left, the 'Connections' panel shows a connection to 'localhost:27017' with a database named 'CinemaDB'. The 'Collections' list on the left includes 'Comments', 'Movies', 'Sessions', 'Theaters', and 'Users'. The 'Theaters' collection is selected. The main panel shows the 'Documents' tab with a query: `select count(theaterId), zipcode from theaters where state = "DC"`. The query is executed, and the results are displayed as three documents. Each document represents a theater in DC, with its location and zipcode.

Document	theaterId	zipcode
1	1092	20010
2	801	20010
3	8500	20010

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

CinemaDB

Comments

Movies

Sessions

Theaters

Users

admin

config

local

My Queries

Users

Theaters

localhost...

Cinem...

Movies

admin

My Qu...

Data Mo...

localhost:27017 > CinemaDB > Theaters

Open MongoDB shell

Documents 1.6K

Aggregations

Schema

Indexes 1

Validation

\$match

\$group

\$project

Edit

Explain

Run

Options

ALL RESULTS

Showing 1 - 3 count results

countTheaterId : 1
zipcode : "20016"

countTheaterId : 1
zipcode : "20010"

countTheaterId : 1
zipcode : "20002"

Nivell 3

Exercici 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

He generado el código con la IA integrada a partir del texto en catalán del ejercicio para comprobar si se puede utilizar otro idioma. La consulta ha devuelto 4 registros que cumplen con las 2 condiciones.

The screenshot shows the MongoDB Compass application running on localhost:27017, connected to the CinemaDB database. The 'Movies' collection is selected. A query filter is applied: `{directors:'John Landis','imdb.rating':{$gte:7.5,$lte:8}}`. The results are displayed in a table with columns: _id, poster, num_mflix_comments, released, awards, countries, cast, directors, runtime, and plot. Two documents are shown:

_id	poster	num_mflix_comments	released	awards	countries	cast	directors	runtime	plot
"573a1397f29313caabce76f7"		1	July 28, 1978 at 12:00:00 AM UTC	[Object]	[Array]	[Array]	[Array]	189	Jake Blues, just out from prison, puts together his old band to save T..."
"573a1397f29313caabce76f7"		1	July 28, 1978 at 12:00:00 AM UTC	[Object]	[Array]	[Array]	[Array]	189	Jake Blues, just out from prison, puts together his old band to save T..."

Nivell 3 - Exercici 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.

En la collection Theaters - Schema hay la opción de hacer un análisis del schema por diferentes criterios. Al contener la collection información sobre las coordenadas geográficas, MongoDB ha situado durante la análisis iniciada las coordenadas en el mapa.

